Cognitive Complexity Theory in HCI

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What is Cognitive Complexity Theory ?

- Cognitive Complexity Theory is a psychological framework used to understand how individuals perceive and try to understand the world.
- This theory proposes that cognitive complexity is the ability to distinguish different aspects of a situation and make them into a coherent whole.
- It is considered a fundamental factor that determines the way individuals process information and make decisions.

- According to Cognitive Complexity Theory, individuals with higher cognitive complexity are better able to process information because they are able to distinguish different aspects of a situation and integrate them.
- In addition, they are better able to deal with ambiguity because they are able to maintain different perspectives and consider different possibilities.

- Cognitive complexity is considered to evolve over time; Because as individuals encounter new situations and ideas, they are forced to reconcile conflicting information and integrate new perspectives.
- This, in turn, can lead to increased cognitive complexity.

- Cognitive Complexity Theory has been applied in many fields.

- In education, it has been used to develop instructional strategies that promote cognitive complexity, such as encouraging students to consider different perspectives and think critically about complex issues.

- In the business world, it has been used to develop training programs to help develop the cognitive complexity necessary to navigate complex and uncertain environments.

General Terms

1.Cognitive Load

Cognitive load refers to the amount that is loaded onto a person's mental resources when performing a task.

Optimizing the design to reduce cognitive load can be an important way to improve the user experience.

cognitive load

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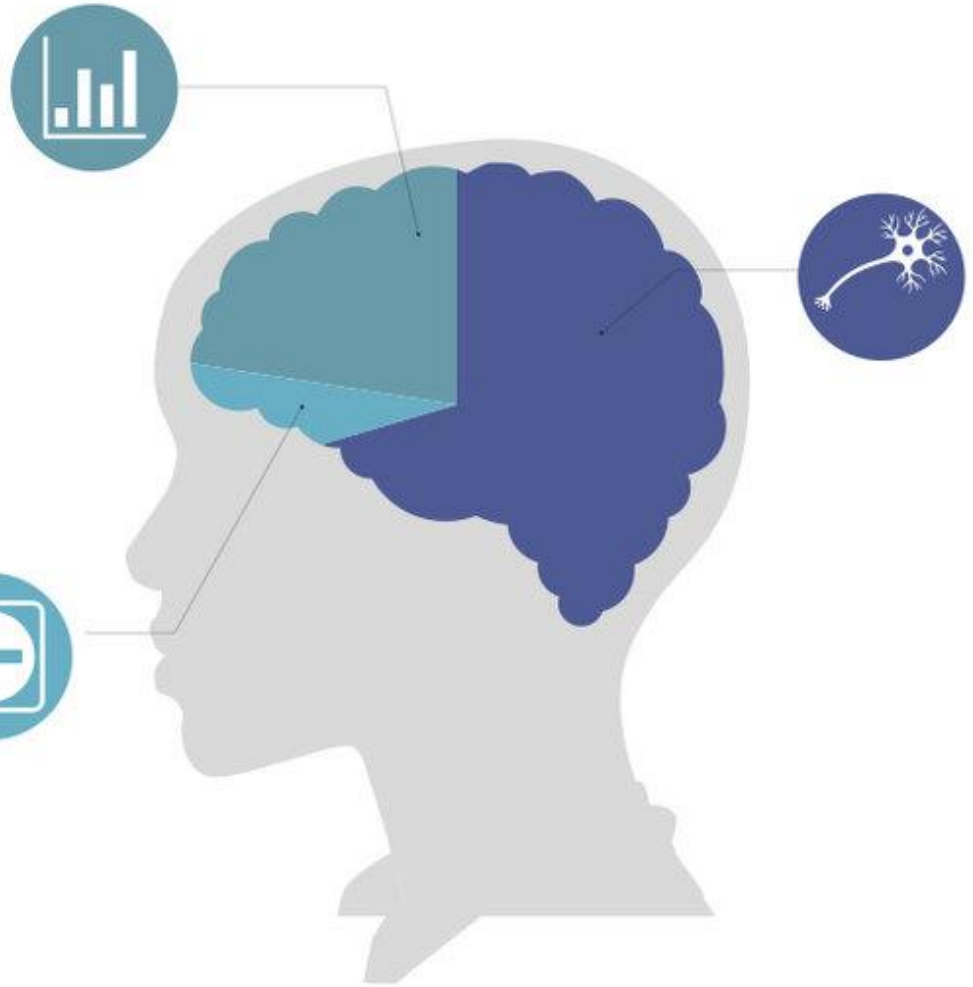


intrinsic cognitive load

complexity of new information

SIMPLIFY

germane cognitive load

deep processing of new information by integrating it with previous learning

MAXIMISE

extraneous cognitive load

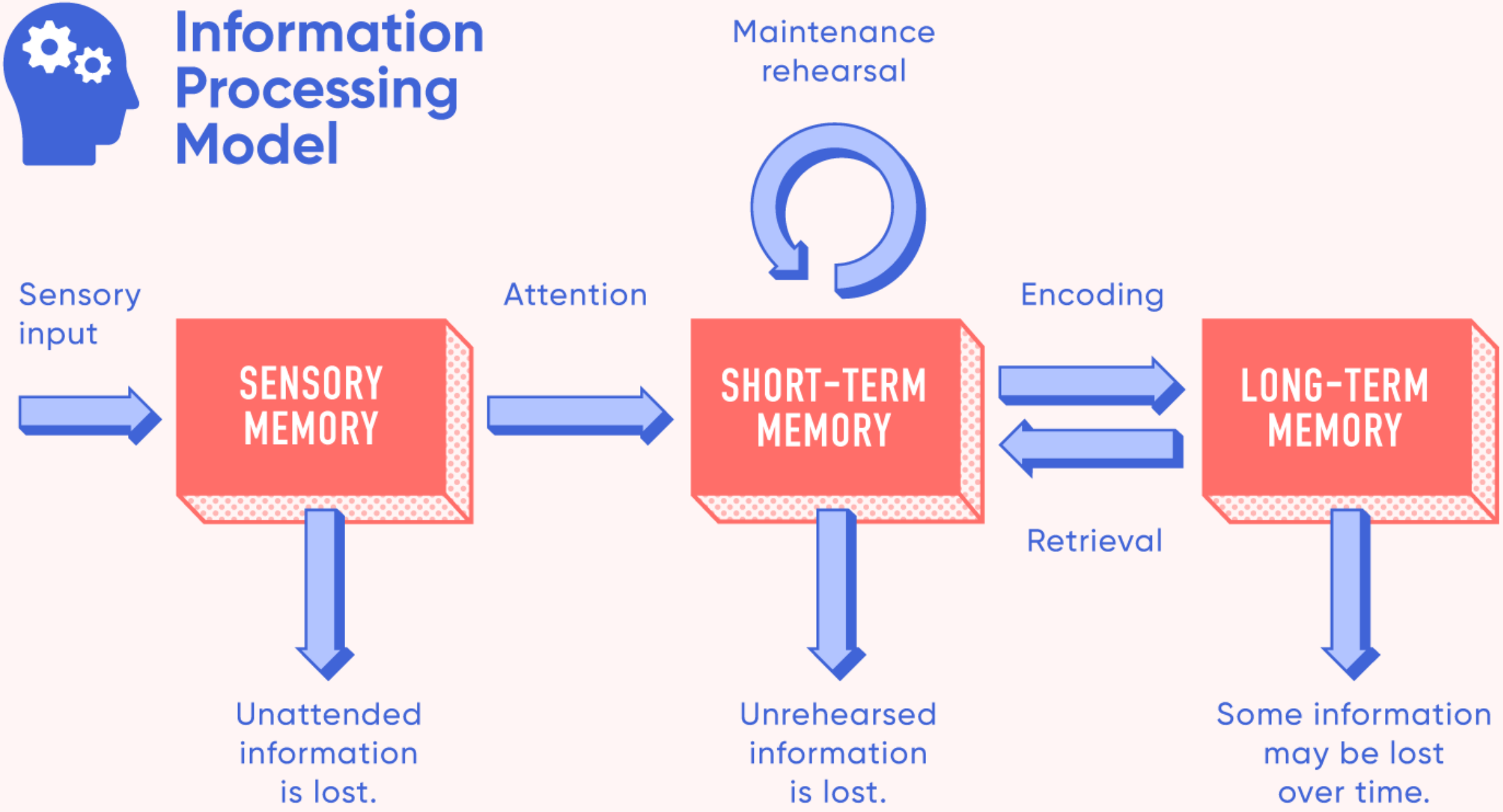
distracts working memory from processing new information

REDUCE

2.Information Processing

- Cognitive Complexity Theory is used to understand information processing processes.
- How individuals receive and process information and make decisions is one of the main focuses of this theory.
- Information processing is important in user interface design in terms of presenting data and the way users understand information.

Information Processing Model

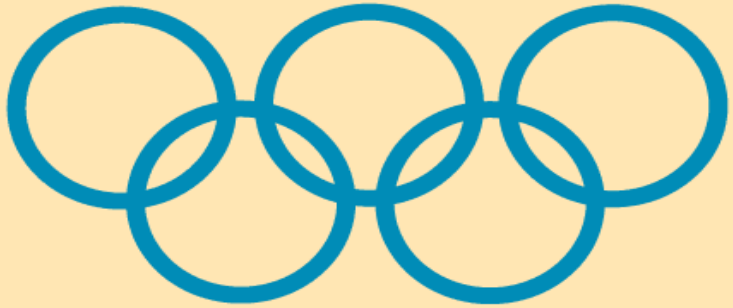


3.Perceptual Organization

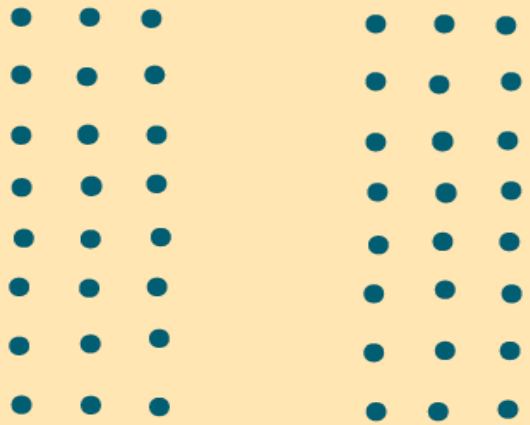
- CCT uses the concept of perceptual organization to understand how users perceive information. This involves the process by which a user understands and organizes information. This plays an important role in the user interface when making design decisions about how information is presented and organized.
- Perceptual organization refers to the way information is received by our senses and interpreted to make it meaningful. A lot of what we know about perceptual organization comes from Gestalt psychology. Gestalt psychologists believe that the human brain perceives a stimulus as a whole and not as the sum of its parts.

Examples of the Gestalt Laws

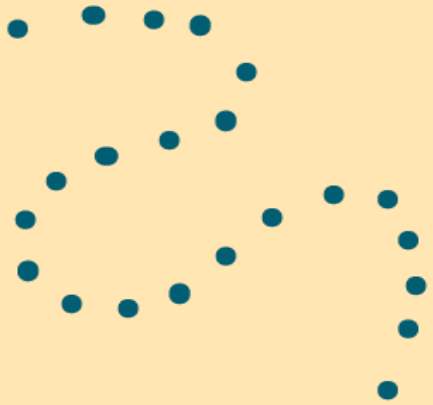
Law of Similarity



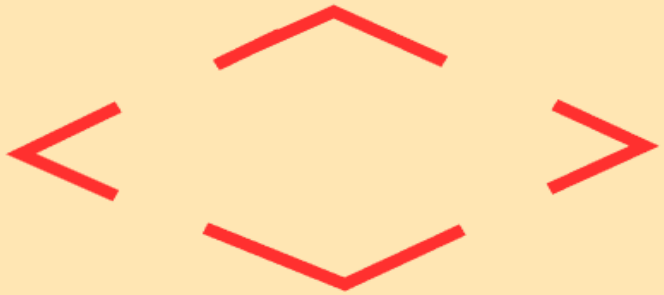
Law of Pragnanz or the Law of Good Figure



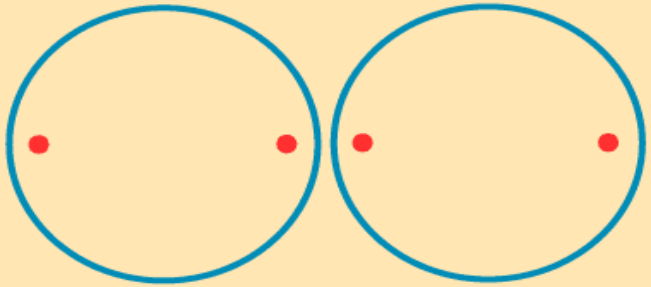
Law of Proximity



Law of Continuity



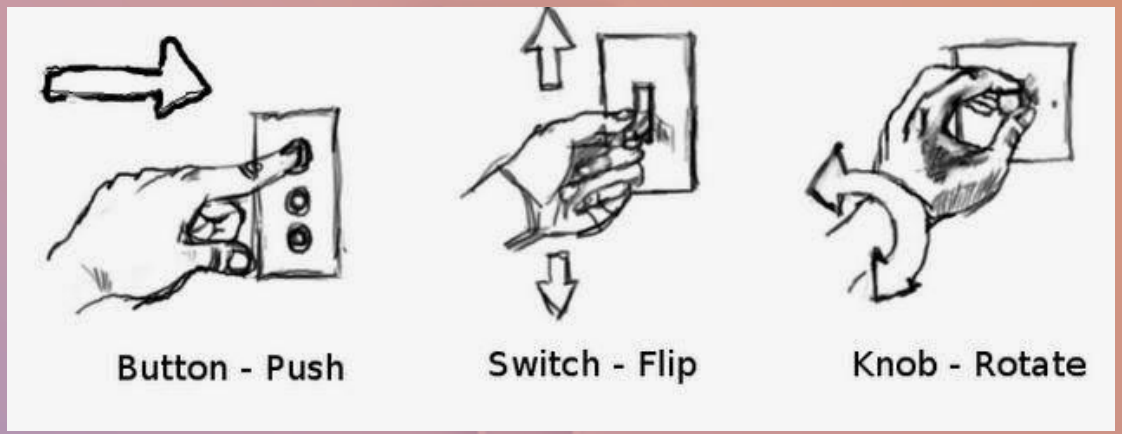
Law of Closure

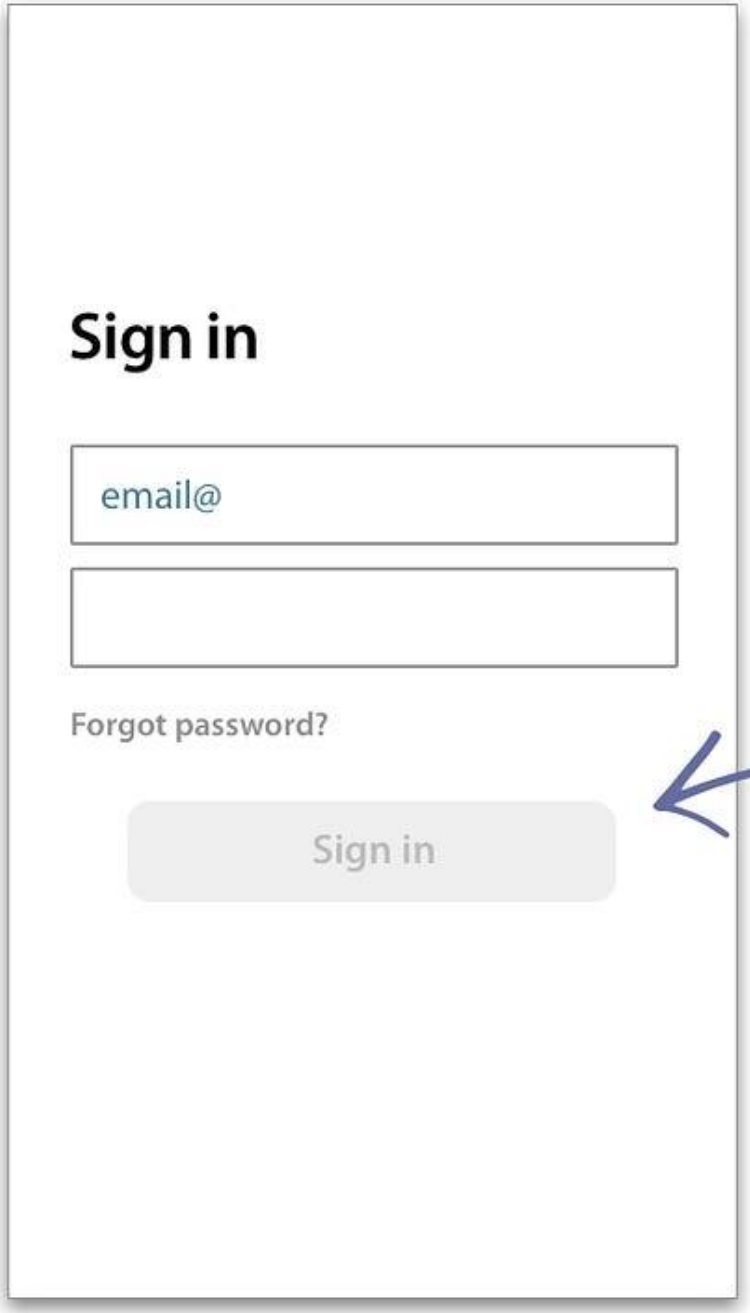


The Law of Common Region

4.Affordances

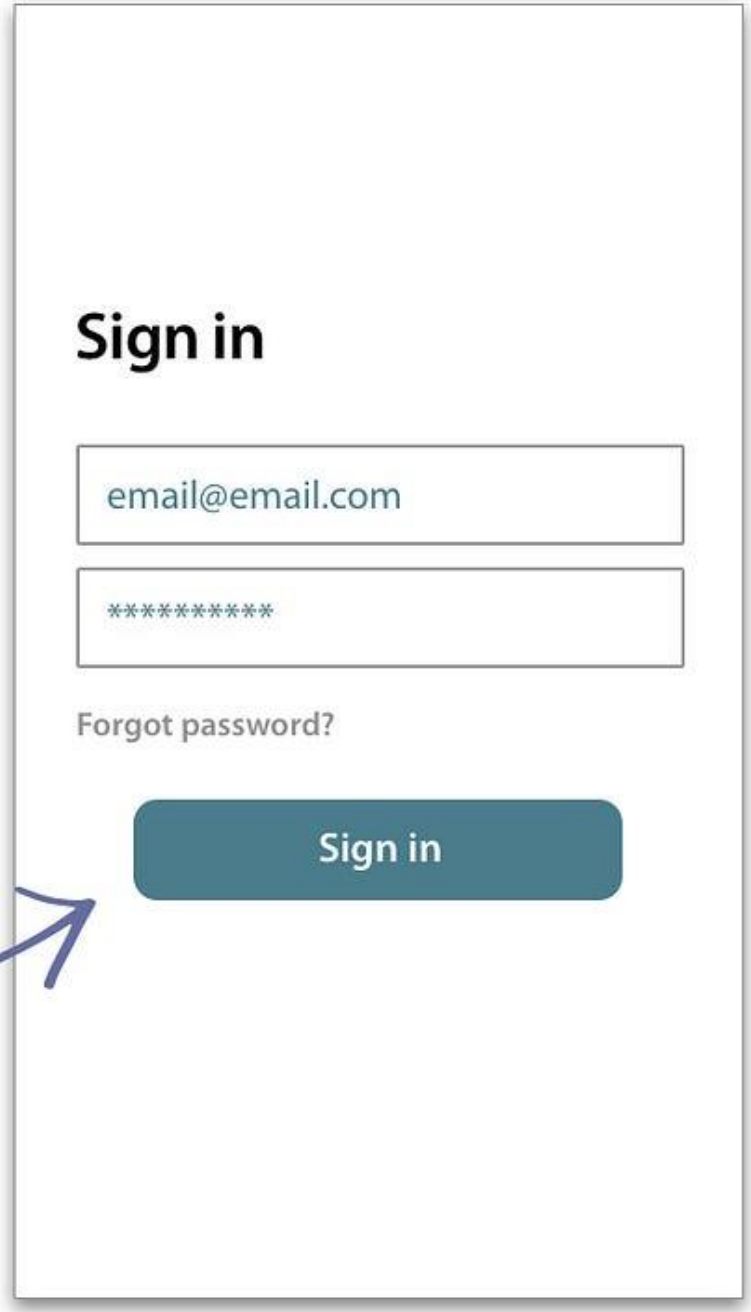
- Cognitive Complexity Theory considers the concept of possibilities in the process of understanding how users can use an object or interface element. Facilities can help users understand how to use something faster. Designers can create user-friendly interfaces by considering the possibilities.





Negative affordance..

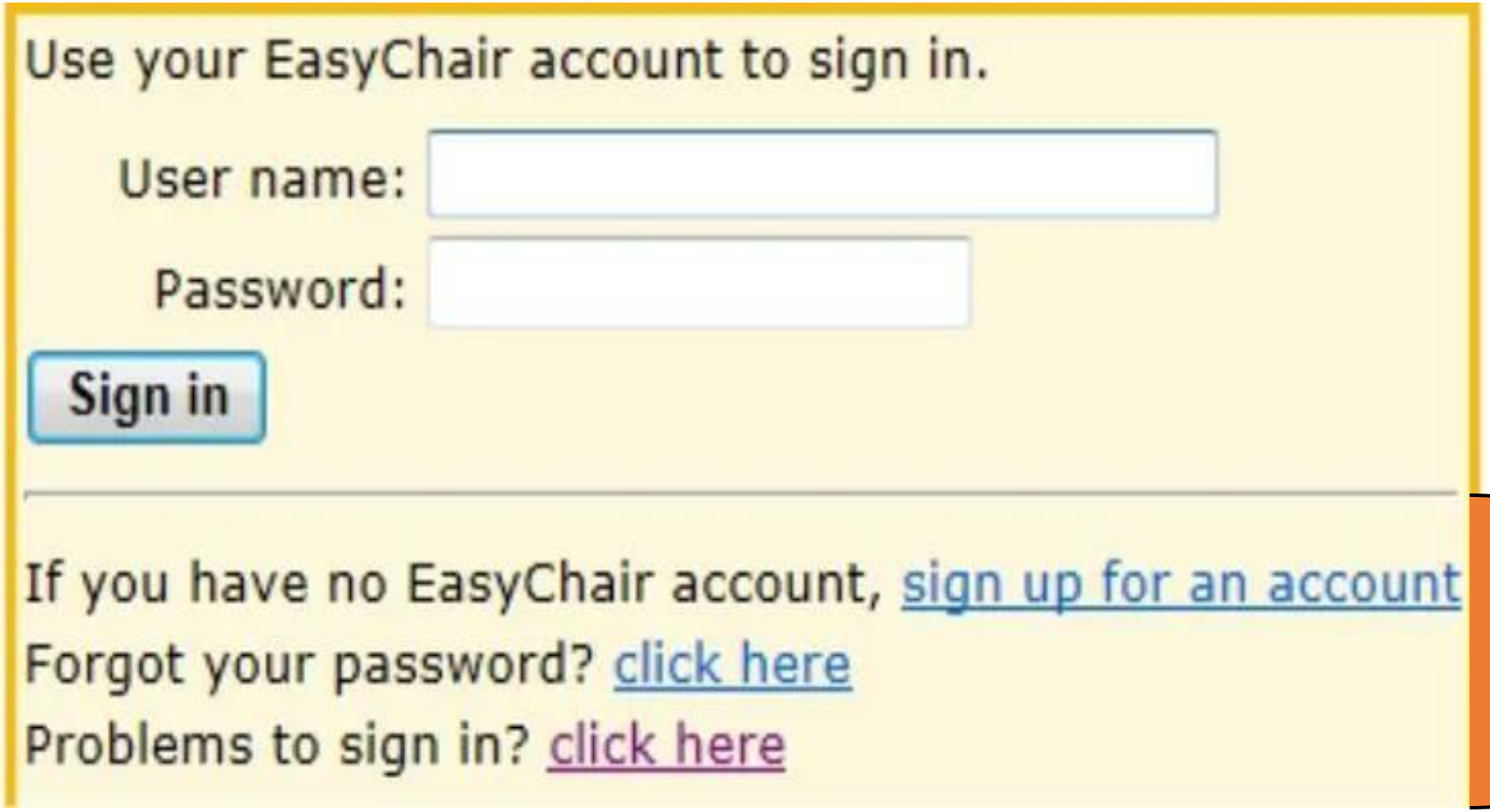
until form is completed



5.Feedback and Feedforward

- CCT emphasizes how users process information through feedback and forward feedback.
- Feedback shows users the results of their actions, while forward feedback provides users with information about their future actions.
- These two concepts can be used in design decisions to enrich the user experience.

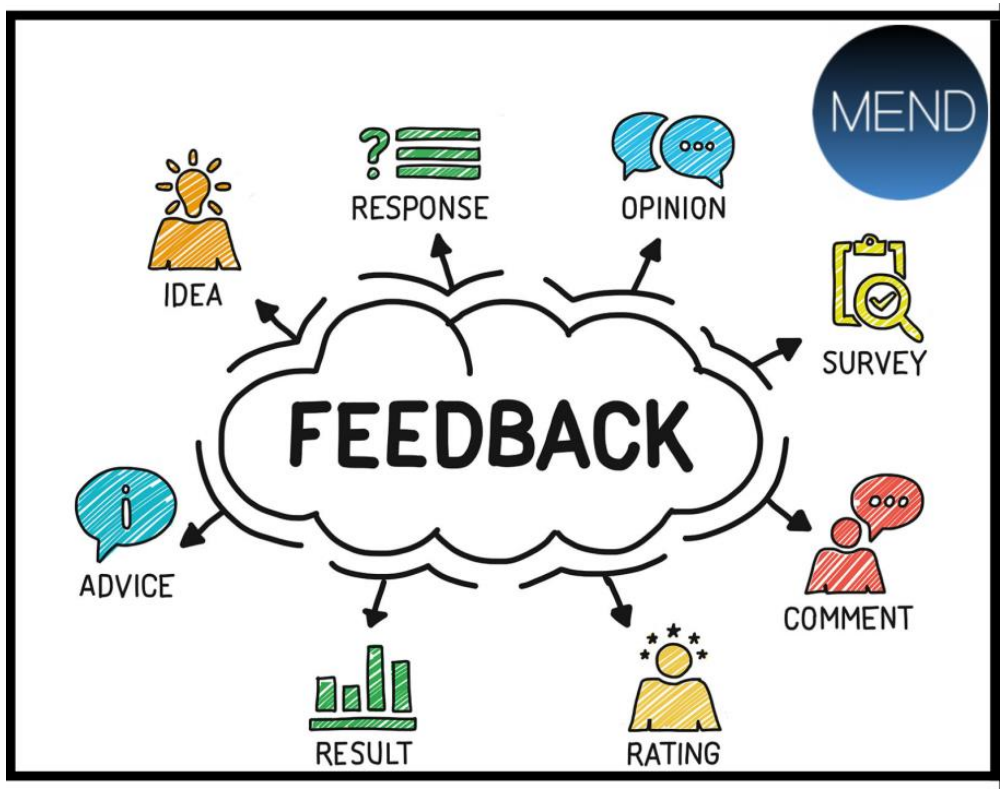
Feedback and Feedforward



Example for
Feedforward

Feedback and Feedforward

- **Feedback, on the other hand, is about what was right and wrong about that job after a completed task or job, and how to do it when this job is repeated.**



6.Skeuomorphism and Flat Design

- Skeuomorphism and flat design are design approaches to how user interfaces are designed.
- Skeuomorphism includes design elements that mimic real-world objects, while flat design offers a more minimalist and modern approach.
- CCT can assess the effects of these design approaches on how users perceive and process information.

iOS 6
Skeuomorphic UI

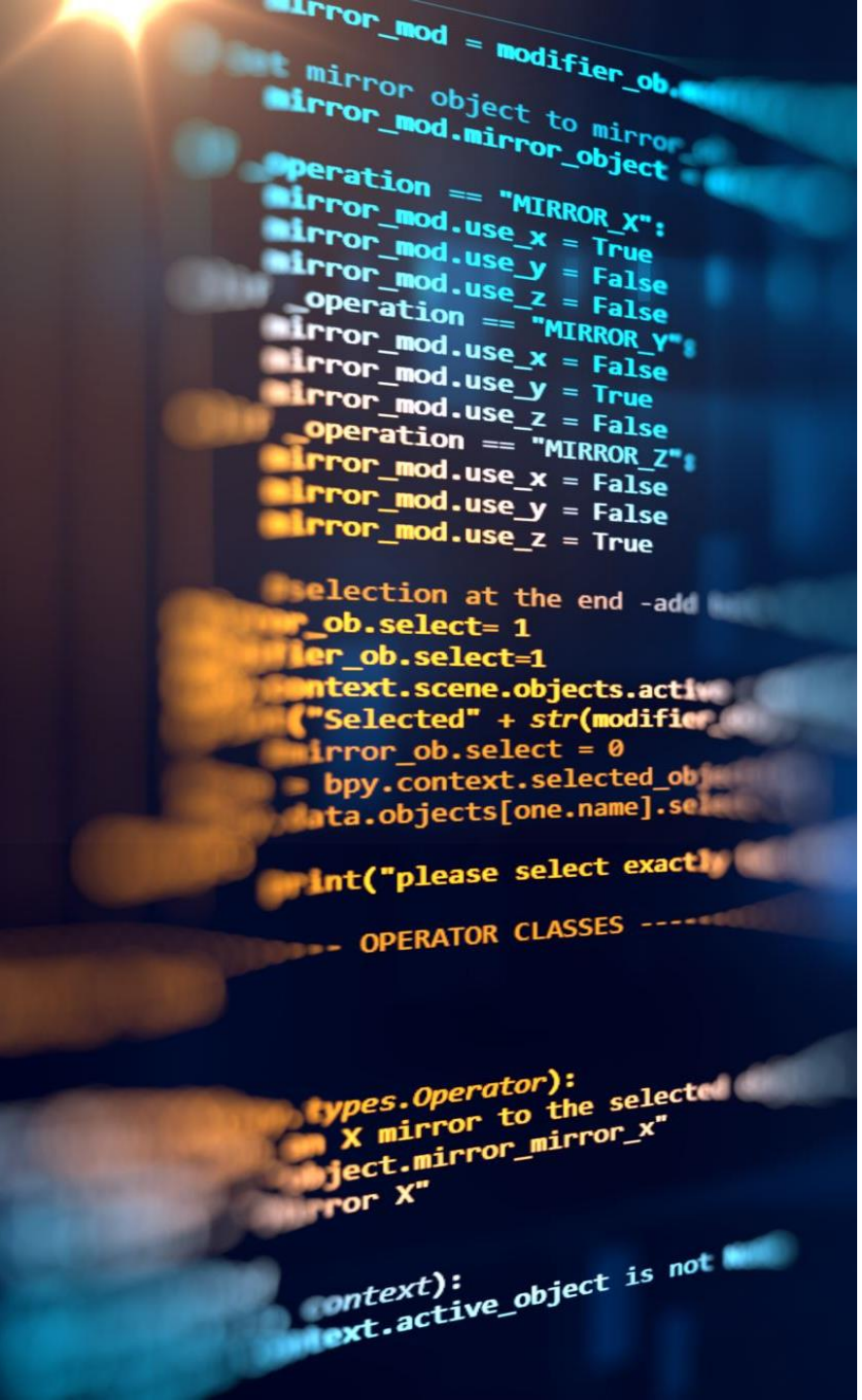
iOS 7
Flat UI

LITERATURE REVIEW

- Article: Research and practice of flipped classroom based on mobile applications in local universities from the perspective of self-determination theory
- Yan’e Hao1*and Yongqiang Lan - 9 January 2023
- URL: https://www.researchgate.net/publication/366989439_Research_and_practice_of_flipped_classroom_based_on_mobile_applications_in_local_universities_from_the_perspective_of_self-determination_theory

- This article discusses how mobile apps and the Flipped Classroom model can be used in education, how they can meet students' basic needs, and how these approaches can affect classroom satisfaction. In addition, it is mentioned that it plays an important role in increasing students' motivation and personalizing their learning experience within the framework of Self-Determination Theory (SDT).
- He also emphasizes the relationship between basic psychological needs and classroom satisfaction. It is proposed to develop such teaching models in order to increase classroom satisfaction by meeting the basic psychological needs of students.

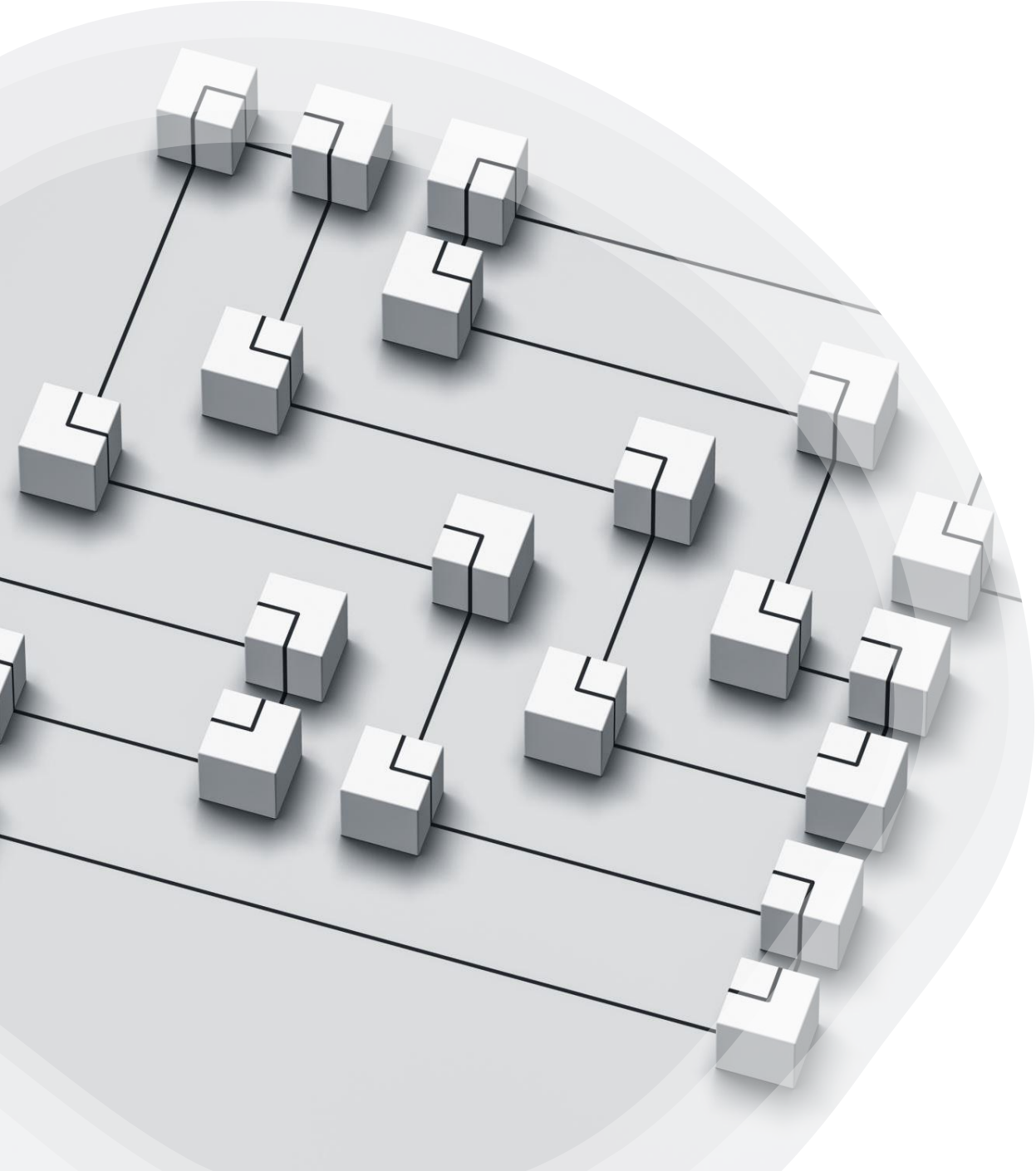
- Article: Applying Cognitive Complexity to Checklist-Based Human-Machine Pair InspectionYan’e Hao1*and Yongqiang Lan - 9 January 2023
- Yujun Dai; Shaoying Liu - 01 April 2022
- URL : https://ieeexplore.ieee.org/document/9742155



- This article emphasizes the rapid growth of software systems and the increasing importance of software quality. Software auditing focuses on the practice of assuring software quality. However, while current auditing techniques often support a peer review approach, this approach may be insufficient for complex systems." It proposes a new approach called "Human-Machine Pair Control". This approach adopts a style in which the machine guides the programmer in checking the code during software development. To achieve this, he uses the theory of Cognitive Complexity to determine how the programmer should supervise. An example application is presented to demonstrate the use, accuracy and usefulness of this method. The results show that this method can help programmers automatically and effectively detect errors during software development.

- Article: How to model behavioural and cognitive complexity in humancomputer interaction with Petri nets
- Rauterberg, G. W. M., Schluep, S., & Fjeld, M - 1 January 1997
- URL : https://ieeexplore.ieee.org/document/647003

- This article summarizes the theory of cognitive complexity in software and the work done on human-computer interaction. The study notes that the decisions and actions that users perform while using the computer are imbued with information that includes mental models, individual problem-solving strategies, and basic decision structure for a given task.



- A tool called AMME analyzes the observed processes and automatically extracts a Petri net description of the task-related decision structure. This network is extended and executed with targeting structures. The goal is to provide functional equivalence between the observed and simulated processes. The article presents three modeling strategies: event-driven, parallel goal setting, and feedback-controlled goal setting.

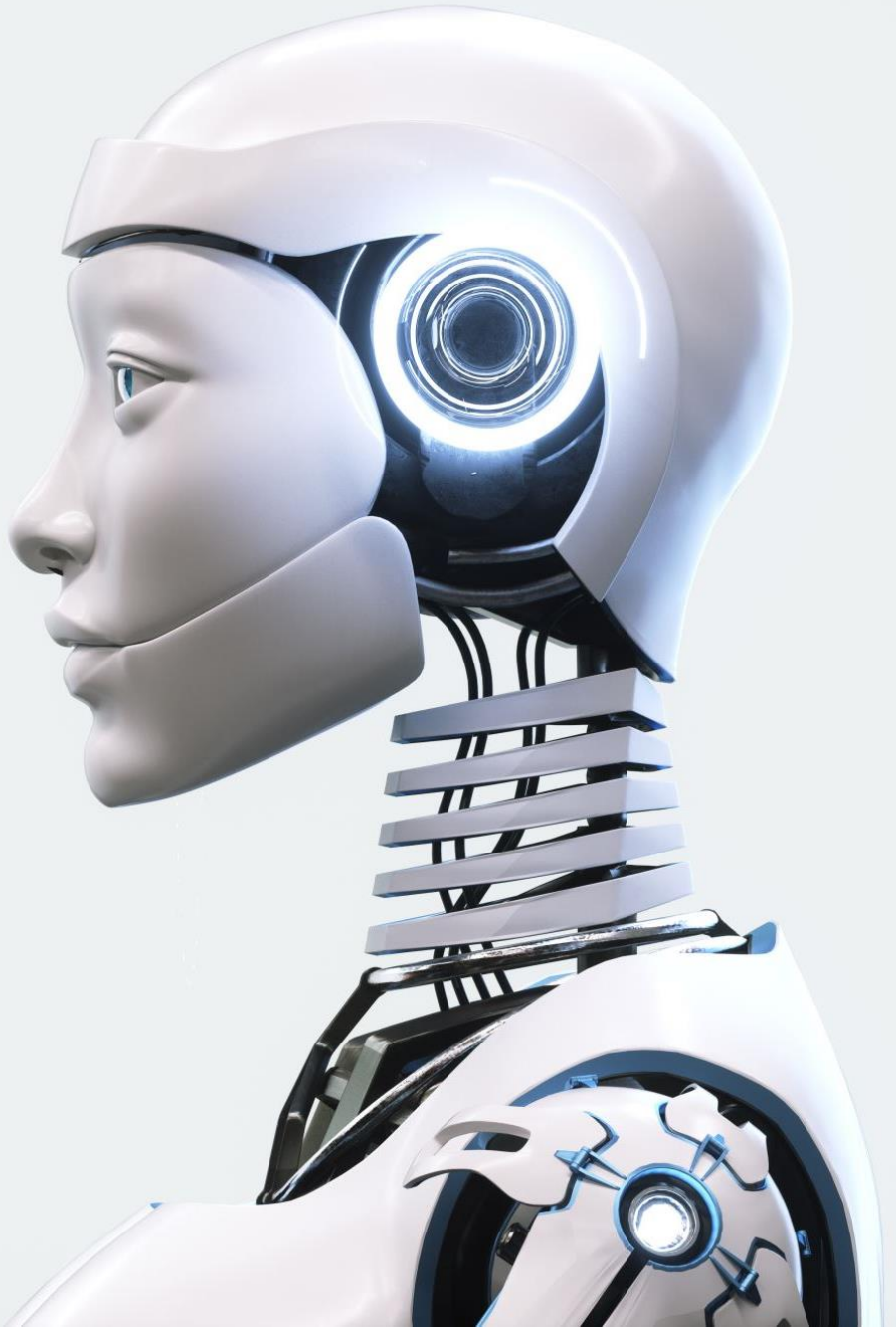
- A simulation study was used to validate these different goal-setting strategies. The simulation results show that more complex mental models provide a better functional equivalence. It was concluded that the feedback-controlled goal-setting strategy combines the advantages of the other two strategies.

- Article: Computers in Human Behavior
- Nina Hollender a,, Cristian Hofmann b, Bernhard Schmitz , Michael Deneke a - 29 June 2010
- URL: https://www.sciencedirect.com/science/article/pii/S0747563210001718?fr=RR-2&ref=pdf_download&rr=81906a4d2ce899f4

- This article focuses on the principles of Cognitive Load Theory (CLT) and instructional design. CLT is a long-used theory in the field of learning and teaching and is based on the concepts of working memory and long-term memory. At first, this theory aimed to reduce the unnecessary cognitive load resulting from the sub-optimal design of learning tasks, but subsequent studies have included teaching principles related to levels of expertise and promoting germanic cognitive load. In addition, the design principles of user experience and human-computer interaction are important for developing complex learning software.

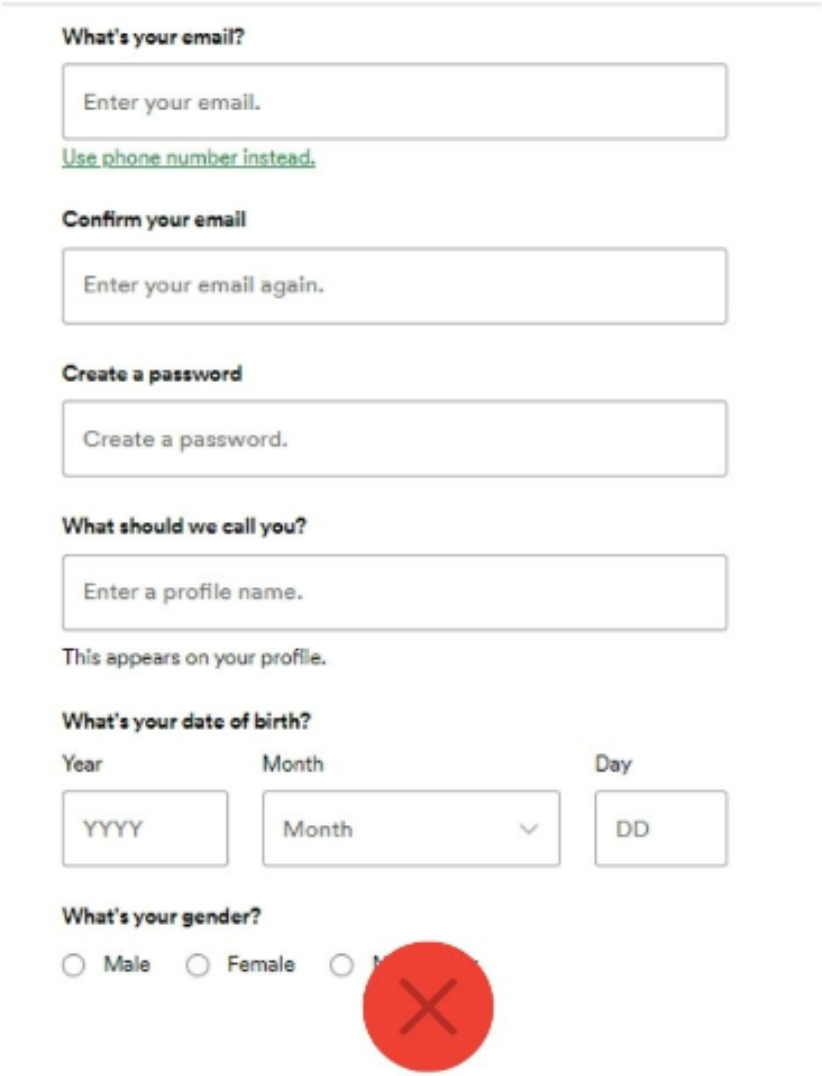
- Article: Mind Meets Machine: Towards a Cognitive Science of Human–Machine Interactions
- Emily S. Cross and Richard Ramsey - 29 June 2015
- URL: https://www.sciencedirect.com/science/article/pii/S1364661320302977?ref=pdf_download&fr=RR-2&rr=81906b8ce90a99f4

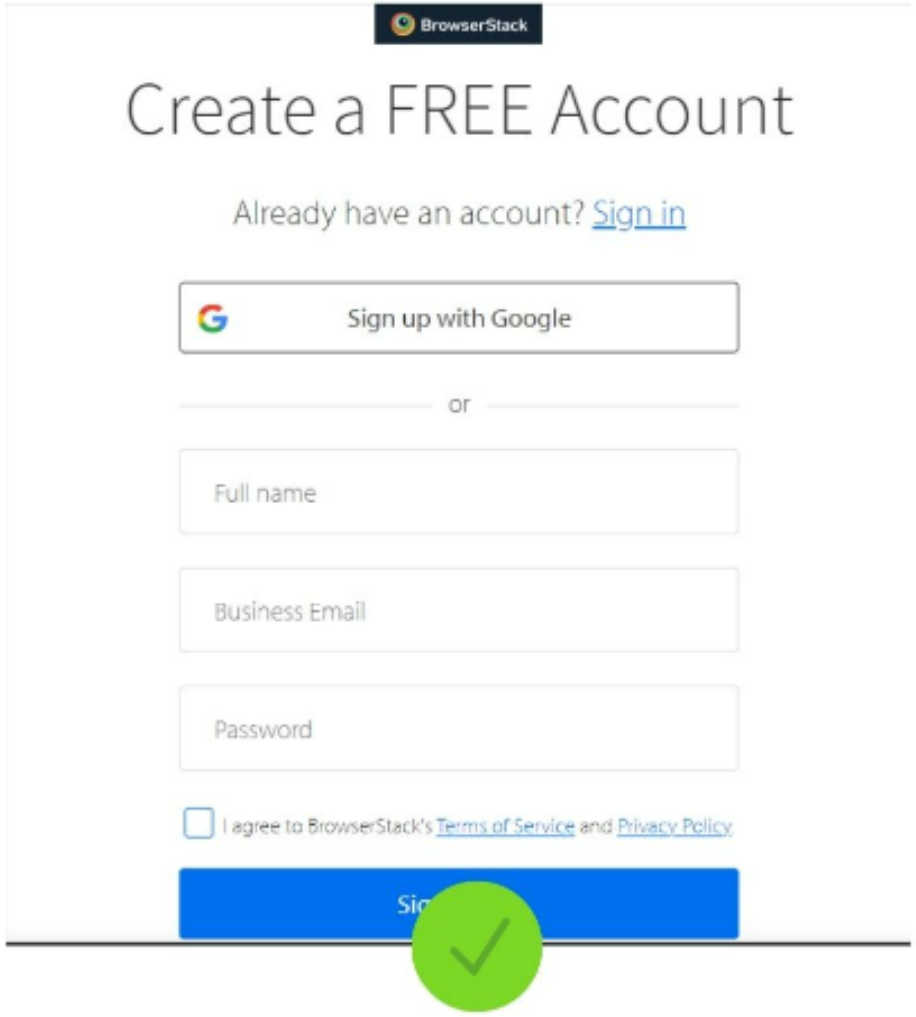
- This article discusses the necessity of understanding the cognitive and brain mechanisms that underpin human-machine interaction at a time when robots and artificial intelligence machines are increasingly taking on social roles in various areas of our lives.



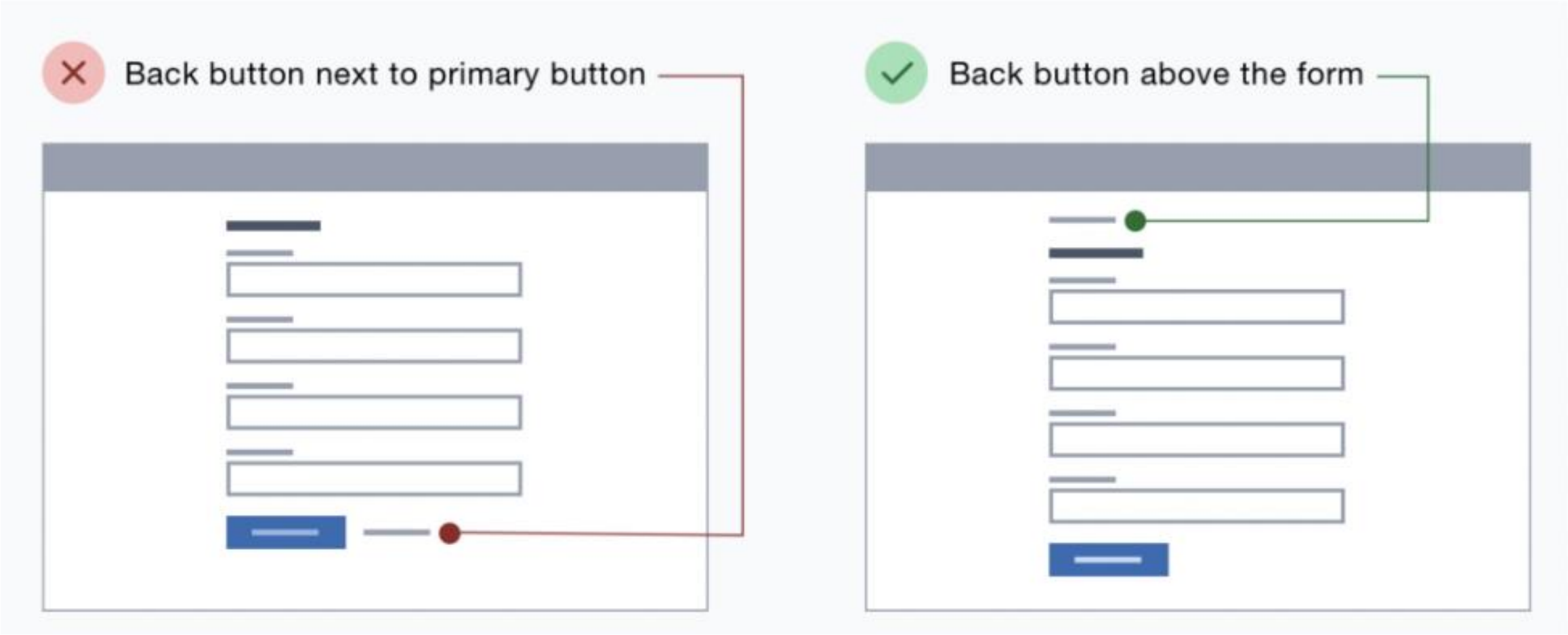
- The authors note that this framework, which deals with human-robot interaction, takes an approach that takes into account the diversity of various machines, the expectations and experiences of individuals, and the structure and function of the mental and brain systems involved in these interactions. This approach encourages interdisciplinary collaboration to better understand the complexity of human-machine interactions. In summary, the text emphasizes the importance of the contribution of cognitive and brain sciences to the study of human-machine interactions, especially as machines incorporating artificial social intelligence become widespread.

Reduce Cognitive Overload





- Consider filling out a signup form on a website. The UI has to be carried out with a few simple steps rather than collecting extensive information at the beginning.
- Look for elements that make the users perform additional tasks such as re-entering details, reading unnecessary content, remembering details, etc. The best alternatives could be displayed as a picture, show previously entered data again (autofill), or establish smart defaults (third party services).


Back button next to primary button
Back button above the form

The minimize, resize, and close buttons will have regular symbols.

The back and forward buttons should have pointing arrows.

Placement of back, cancel buttons, etc.

Bad
Sign up Form

Better
Sign up Form
Group 2
Group 3

- For those of us in the design field, Miller’s Law offers a valuable lesson: the technique of “chunking.” By organizing content into smaller, more digestible groups, we make it easier for users to process and retain information. This is especially crucial for first-time users who haven’t yet committed the information to long-term memory.

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